Activity 1.2 – 8-bit Up/Down Counter

1. Design Approach:

Designed an 8-bit synchronous up/down counter using Verilog HDL. It supports synchronous reset, enable, up/down control and parallel loading. Reset has the highest priority, followed by load and enable.

2. Verification Steps:

- Created a separate testbench.
- Tested synchronous reset.
- Checked parallel loading.
- Verified counting up and counting down.
- Tested enable and hold operations.
- Checked overflow and underflow conditions.
- Simulated using ModelSim.

3. Results:

The testbench checks the counter output after every clock edge and displays PASS/FAIL messages.

4. Issues Faced:

Correct clock-edge operation and priority between reset, load and enable must be maintained.

5. Conclusion:

Successfully designed and verified an 8-bit synchronous up/down counter.